

Jack Rose

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EDUCATION

Ohio State University

B.S. in Data Analytics, Philosophy Minor

Coursework: Algorithms, Linear Algebra, AI/ML, Statistics, Optimization.

Expected May 2027

Cumulative GPA: 3.51

TECHNICAL SKILLS

Languages	C/C++, Python, R, Java, JavaScript
Systems/Performance	SIMD (AVX2), Multithreading, Memory Management, Low-Level Optimization
Data & Analytics	SQL, Pandas, NumPy, Neural Networks, R, Statistical Modeling
Optimization	Gurobi, Linear Programming, MILP

ENGINEERING PROJECTS

Deepity — SIMD-Accelerated Predictive Coding Library C++20, AVX2/AVX-512, OpenMP, pybind11 [GitHub](#)

- Built a predictive coding neural network library from scratch in bare-metal C++ with Python bindings, reaching 97.73% MNIST accuracy in 60 seconds on a laptop CPU — within 1% of PyTorch backpropagation on the same architecture.
- Hand-wrote AVX2/AVX-512 activation kernels, a cache-aligned memory arena, and an exact mu-caching optimization, sustaining ~123 GFLOPS and running 3.6x faster than a NumPy implementation.

Urban Facility MILP Optimizer Python, Gurobi [GitHub](#)

- Formulated a Mixed-Integer Linear Programming (MILP) model to optimize spatial facility placement against strict maximum Manhattan distance constraints. Engineered the algorithm to process 10.6 million simplex iterations across 32,902 nodes in under 130 seconds.

Local AI Content Moderation Engine Node.js, DistilBERT, ONNX [GitHub](#)

- Deployed a 66M parameter DistilBERT neural network into a local Node.js backend using the ONNX Runtime to eliminate API latency.
- Incorporated a lighter, Logistic Regression model to Nexirift, a social media startup.

RESEARCH EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

Department of Radiation Oncology, Ohio State University College of Medicine | Columbus, OH

- Contribute to evaluation and development of clinical large language models (LLMs) and vision-language models (VLMs).
- Work with open-source AI frameworks including PyTorch and Hugging Face.
- Assist in benchmarking, model evaluation, and analysis of healthcare-oriented AI systems.

Epistemology & Ethics Research

Jan 2026

Sorge Philosophical Journal

- Presented original research, “Truth is Conditional,” at the Ohio State Union Symposium (2026), proposing an epistemic utility framework for rational disagreement based on conditional probability and background-state dependence.
- Published in Sorge, Ohio State’s undergraduate philosophy journal; nominated for the Undergraduate Bingham Award.